

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Case Study

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA15
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COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Dave's Taxonomy: Precision (Level 3)
Question Count: 3

Total Marks: 30

Code of Conduct

I _Narmatha J (192521341)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: __Narmatha J__

Assessment Tasks

Case Study

1. Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

2. Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

3. Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized

ANSWERS:

Case Study 1: Smart Retail Inventory Management System

Introduction:

A retail chain has multiple stores located in different places. Managing product stock, sales transactions, and customer demand manually is difficult and can lead to stock shortages or excess inventory. To solve this problem, the company implements a **cloud-based Smart Retail Inventory Management System**.

The system connects Point-of-Sale (POS) machines, store devices, dashboards, and centralized cloud services. The cloud platform provides storage, analytics, demand forecasting, and secure communication between different components.

System Architecture:

The system consists of several important cloud components:

- **Client Layer:** POS systems and management dashboards are used by store employees and administrators.
- **API Layer:** APIs allow POS systems and dashboards to communicate with cloud services.
- **Cloud Storage:** Stores product information, sales transactions, inventory records, and analytics data.
- **Platform Services:** Cloud-based analytics and demand forecasting services analyze sales patterns.
- **Infrastructure:** Cloud servers provide the required computing and networking resources.
- **Security Layer:** VPNs, firewalls, authentication mechanisms, and API gateways protect the system.

Working of the System:

When a customer purchases a product, the POS system records the transaction. This information is sent to the centralized cloud system through secure APIs.

The cloud application updates the inventory database immediately. Managers can access dashboards to check the current stock levels of different stores.

The analytics service processes historical sales information and identifies purchasing patterns. The demand forecasting service predicts which products are likely to be required in the future. Based on these predictions, the retail chain can reorder products before they become unavailable.

Security:

Security is an important part of the system because it handles business and transaction data.

The system uses:

- **VPN:** Provides secure communication between stores and cloud services.
- **Firewall:** Blocks unauthorized network traffic.
- **API Gateway:** Controls and monitors API requests.
- **Authentication:** Ensures that only authorized users can access the system.
- **Encryption:** Protects sensitive information during transmission and storage.

Benefits:

The Smart Retail Inventory Management System provides several benefits:

1. Real-time monitoring of inventory.
2. Reduces stock shortages and overstocking.
3. Helps predict future product demand.
4. Provides centralized access to information.
5. Improves decision-making through analytics.
6. Reduces manual inventory management.
7. Provides secure communication between stores and cloud services.
8. Can easily scale when new stores are added.

Conclusion:

The Smart Retail Inventory Management System demonstrates how **cloud infrastructure, platform services, APIs, storage, and security mechanisms** can work together. By using cloud-based

analytics and demand forecasting, the retail chain can improve inventory accuracy, reduce operational costs, and provide better customer service.

Case Study 2: Cloud-Based Document Collaboration System

Introduction:

Organizations often need multiple employees to work on the same documents from different locations. Traditional document-sharing methods can create problems such as duplicate files, outdated versions, and difficulty in tracking changes.

To overcome these problems, an enterprise deploys a **Cloud-Based Document Collaboration System**, similar to Google Docs. Users can access documents through web browsers and collaborate in real time.

System Architecture:

The system contains the following major components:

- **Client Layer:** Users access the system through web browsers, laptops, tablets, and smartphones.
- **Web Application:** Provides the interface for creating, editing, and sharing documents.
- **Web APIs:** Synchronize document changes between different users and devices.
- **Distributed Cloud Storage:** Stores documents and related data across multiple cloud servers.
- **Version Control:** Maintains previous versions of documents.
- **Network Infrastructure:** Provides reliable and low-latency communication.
- **Security Services:** Provide encryption, identity management, and access control.

Working of the System:

A user logs into the collaboration platform through a web browser and opens a document stored in cloud storage.

When the user edits the document, the changes are sent to the cloud through web APIs. The cloud service processes these changes and synchronizes them with other users who are currently viewing the same document.

For example, if three employees are working on the same report, one employee can add text while another modifies a section. The system synchronizes the changes so that all authorized users see the updated document.

The version control mechanism maintains previous versions. If an important change is accidentally deleted, an earlier version can be restored.

Security:

Since business documents may contain confidential information, strong security mechanisms are required.

The system uses:

- **Encryption:** Protects documents during transmission and storage.
- **Identity Management:** Verifies the identity of users.
- **Access Permissions:** Controls who can view or edit documents.
- **Authentication:** Prevents unauthorized users from accessing accounts.
- **Secure APIs:** Protect communication between clients and cloud services.

Benefits:

The document collaboration system provides the following advantages:

1. Enables real-time collaboration.
2. Allows users to work from different locations.
3. Eliminates multiple copies of the same document.
4. Provides automatic document synchronization.
5. Maintains previous document versions.
6. Reduces dependency on physical storage.
7. Provides high availability through distributed cloud infrastructure.

8. Improves productivity and teamwork.

Conclusion:

The Cloud-Based Document Collaboration System demonstrates how **cloud storage, APIs, distributed infrastructure, networking, and security services** can be integrated to provide real-time collaboration. It allows organizations to securely create, edit, share, and manage documents from anywhere.

Case Study 3: Online Food Ordering Platform

Introduction:

Online food ordering has become an important digital service. Customers expect to browse menus, place orders, make payments, and track their orders using mobile applications or websites.

A startup develops a **Cloud-Based Online Food Ordering Platform** to provide these services. The system uses cloud computing to handle a large number of users and orders while maintaining reliability and scalability.

System Architecture:

The platform consists of several cloud components:

- **Client Layer:** Web browsers and mobile applications used by customers.
- **Frontend:** Provides the user interface for browsing menus and placing orders.
- **RESTful Web APIs:** Connect frontend applications with backend services.
- **Backend Services:** Process user accounts, menus, orders, and other operations.
- **Cloud Storage:** Stores menu images, invoices, customer information, and order data.
- **Virtual Machines:** Provide computing resources for backend applications.
- **Managed Kubernetes:** Manages containers and automatically supports scaling.
- **Security Services:** Provide authentication, HTTPS, and role-based access control.

Working of the System:

A customer opens the food ordering website or mobile application and browses the available restaurants and menus.

The frontend sends requests to backend services using **RESTful APIs**. When the customer selects food items and places an order, the backend processes the request and stores the order information in cloud storage.

The system can also generate an invoice and store it securely. Restaurant employees can access the orders through authorized accounts.

During peak periods, such as weekends or festivals, the number of users may increase significantly. Virtual machines and managed Kubernetes help the platform increase computing resources and handle the additional workload.

Security:

The platform handles customer and order information, so security is essential.

The system uses:

- **Authentication Tokens:** Verify users when they access services.
- **HTTPS:** Encrypts communication between clients and servers.
- **Role-Based Access Control (RBAC):** Gives users permissions based on their roles.
- **Secure Cloud Storage:** Protects customer and order information.
- **API Security:** Prevents unauthorized access to backend services.

For example, a customer should only be able to view their own orders, while a restaurant administrator can manage menu items and restaurant orders.

Scalability and Reliability:

One of the major advantages of using cloud computing is the ability to handle changing workloads.

When the number of customers increases, additional computing resources can be provided. **Kubernetes** helps manage application containers and distribute workloads efficiently.

If one server experiences a problem, other instances can continue handling requests. This improves the availability and reliability of the food ordering platform.

Benefits:

The cloud-based food ordering system provides several advantages:

1. Supports web and mobile users.
2. Provides real-time order processing.
3. Can handle a large number of users.
4. Provides scalable cloud infrastructure.
5. Reduces the need for physical servers.
6. Provides secure storage for customer information.
7. Improves application reliability.
8. Allows restaurants and customers to access the system from anywhere.

Conclusion:

The Online Food Ordering Platform demonstrates the integration of **cloud infrastructure, RESTful APIs, cloud storage, virtual machines, Kubernetes, and security services**. Cloud computing allows the startup to provide a reliable and scalable service while handling changing numbers of customers and orders.